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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/17/2006 |
| **PI Name:** | Bayorh, Mohamed A Ph.D. | | |
| **Project Title:** | Possible Countermeasures to Post-Suspension Hypotension in the Head-Down Tilt Rat Model | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Cardiovascular Alterations Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | bayorh@msm.edu | **Fax:** | FY 404-752-1164 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 404-752-1714 |
| **Organization Name:** | Morehouse School of Medicine | | |
| **PI Address 1:** | 720 Westview Drive, South West | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Atlanta | **State:** | GA |
| **Zip Code:** | 30310-1495 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | NSBRI |
| **Start Date:** | 05/01/2001 | **End Date:** | 07/31/2004 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NCC 9-58-CA.002.12 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Exposure to microgravity in humans causes cardiovascular deconditioning with orthostatic hypotension and tachycardia. Postflight orthostatic intolerance is a dramatic physiological consequence of human adaptation to microgravity made inappropriate by a sudden return to normal gravity. Loss of appropriate cardiovascular reflexes contributes to the cardiovascular deconditioning, but the specific mechanisms remain unclear. The hypothesis of the proposed studies is that post-suspension hypotension in rats following simulated microgravity is due to overproduction of nitric oxide (NO) and prostacyclin (PGI2). Also, gender and age may be significant factors that have not been adequately evaluated. To test the above hypothesis, we propose to use the 30o tail-suspended (hindlimb-unloaded) rat model to evaluate the roles of gender and age and to evaluate the selective nitric oxide synthase II (NOS II) inhibitor AMT (2-amino-dihydro-6-methyl-4H-1,3-thiazine) as countermeasures against post-suspension hypotension in Sprague-Dawley rats. We will investigate the specific cardiovascular and hemodynamic parameters (i.e., blood pressure and heart rate; blood flow), vascular remodeling (aorta and mesenteric arteries) and signal transduction (i.e., the role of | | |

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| | NO, cGMP, PGI2, TXA2, ? enzyme immunoassays) mechanisms involved, including detailed studies on baroreflex responses and vascular reactivity. Data derived from the proposed studies will lead to a better understanding of the role of gender and age in simulated microgravity. The major objective of this proposal is to seek more effective countermeasures to cardiovascular deconditioning/orthostatic hypotension observed post-flight. The endothelium is now recognized to play a critical role in the regulation of vascular resistance and blood pressure through the release of nitric oxide (NO) and prostacyclin (PGI2). Thus, a greater understanding of the detailed mechanisms (e.g., cardiovascular, signal transduction, etc.) as well as the role of gender and age in the actions of NO and PGI2 in post-suspension hypotension will lead to the development of more effective countermeasures. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The major objective of this proposal is to seek more effective countermeasures to cardiovascular deconditioning/orthostatic hypotension observed post-flight. The endothelium is now recognized to play a critical role in the regulation of vascular resistance and blood pressure through the release of nitric oxide (NO) and prostacyclin (PGI2). Thus, a greater understanding of the detailed mechanisms (e.g., cardiovascular, signal transduction, etc.)as well as the role of gender and age in the actions of NO and PGI2 in post-suspension hypotension will lead to the development of more effective countermeasures. |
| **Task Progress:** | No progress reported this reporting period. |
| **Bibliography Type:** | Description: (Last Updated: 02/27/2004) |